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Learn
ASP.Net Web APIs With .Net 5
From Scratch

...

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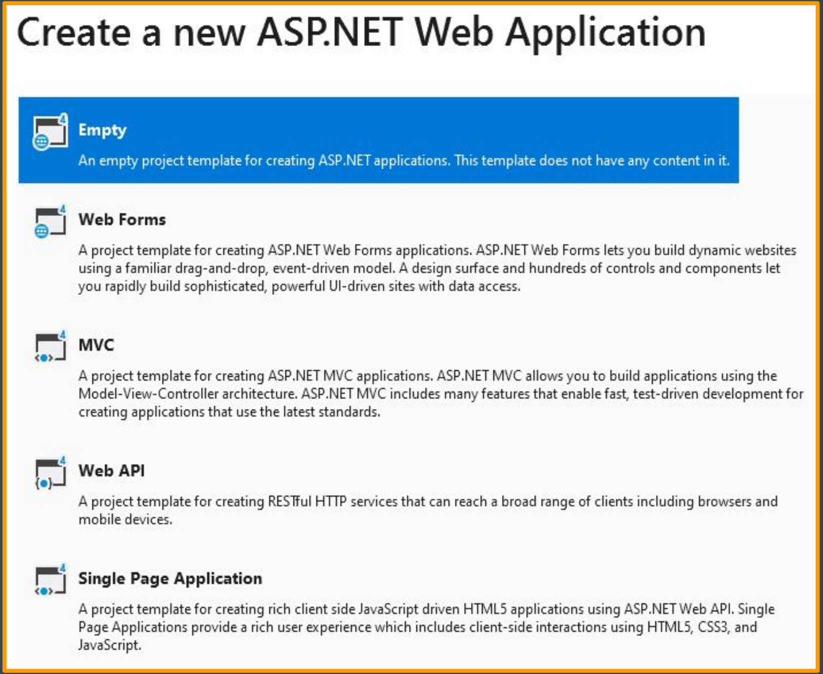
Prerequisite

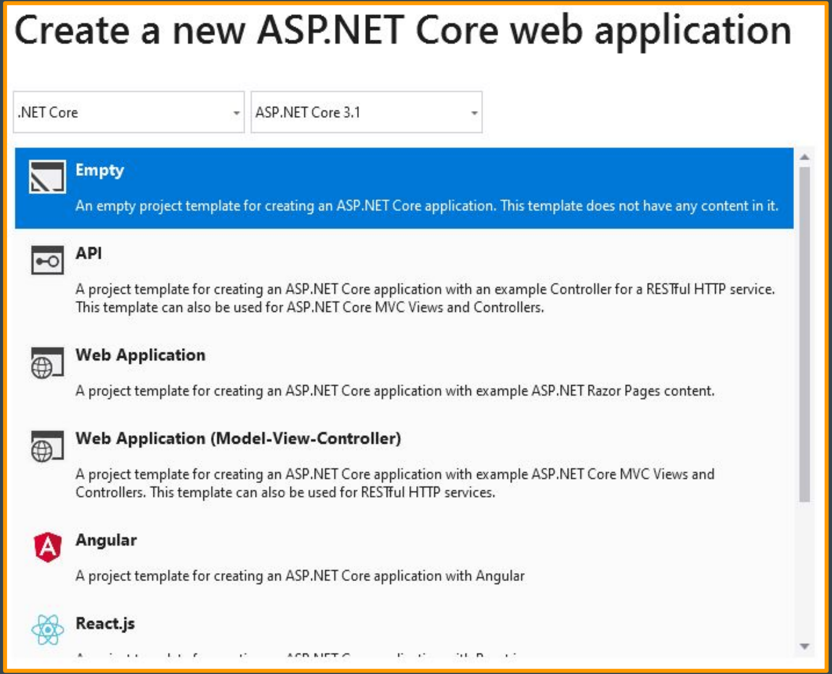
- SQL Server
- C#.Net Core
- EF Core

ASP.Net Core in .Net 5

- ASP.Net is a part of Microsoft.Net for developing web apps.
- ASP.Net Core is the redesign of ASP.Net.
- It is open source.
- It can target Windows, Linux & Mac OS.
- No ASP.Net Web Forms :((ASPX Pages)
- It has 3 major modules
  - MVC (Web Application)
  - Razor Pages (Web Application)
  - Web APIs (Service)

ASP.Net Vs ASP.Net Core With VS 2019





First ASP.Net Core Static Website

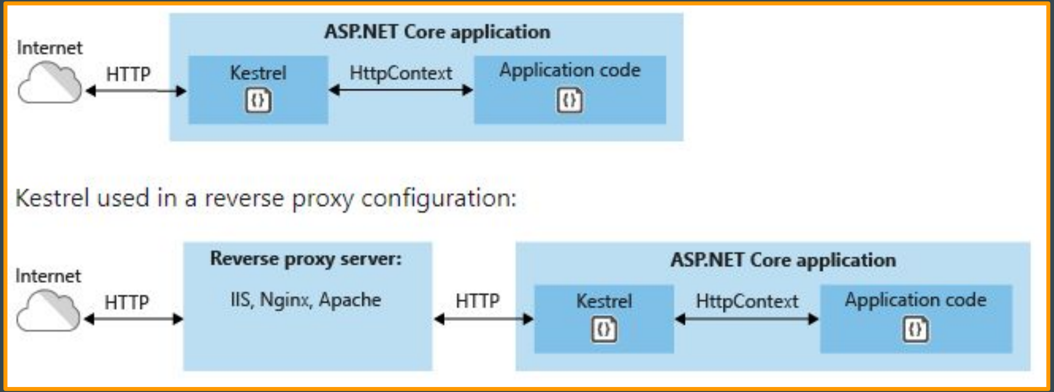
- Design a static website
- Set a default startup page
- https://github.com/damianh/StaticFilesMiddleware/blob/master/src/Microsoft.Owin.StaticFiles/DefaultFilesOptions.cs (for ref)

```csharp
public void Configure(IApplicationBuilder app, IHostingEnvironment env)
{
    app.UseDefaultFiles();
    app.UseStaticFiles();
}
```

```csharp
public void Configure(IApplicationBuilder app, IHostingEnvironment env)
{
    DefaultFilesOptions defaultFilesOptions = new DefaultFilesOptions();
    defaultFilesOptions.DefaultFileNames.Clear();
    defaultFilesOptions.DefaultFileNames.Add("Home.htm");
    app.UseDefaultFiles(defaultFilesOptions);
    app.UseStaticFiles();
}
```

Understanding The Main() Of Program.cs Method

- Understand directory structure
  - CreateDefaultBuilder() : Use Kestrel(cross-platform web server)
  - UseStartup<Startup>() : ConfigureServices(), Configure()
  - Build() : Makes the server ready.
  - Run() : Starts Listing.



Understanding The Startup Class

- Understanding wwwroot Folder
- **ConfigureServices() Vs Configure()**
  - CogifureServices() : Adding services to IServiceCollection container will make them available for dependency injection.
  - Cofigure() : It is a middleware i.e., pipeline to handle requests and responses. Ordering is important in Configure method.

ASP.Net Core Web APIs In Simple Words

It is a program that takes request (as JSON) and send response (as JSON) for any UI.

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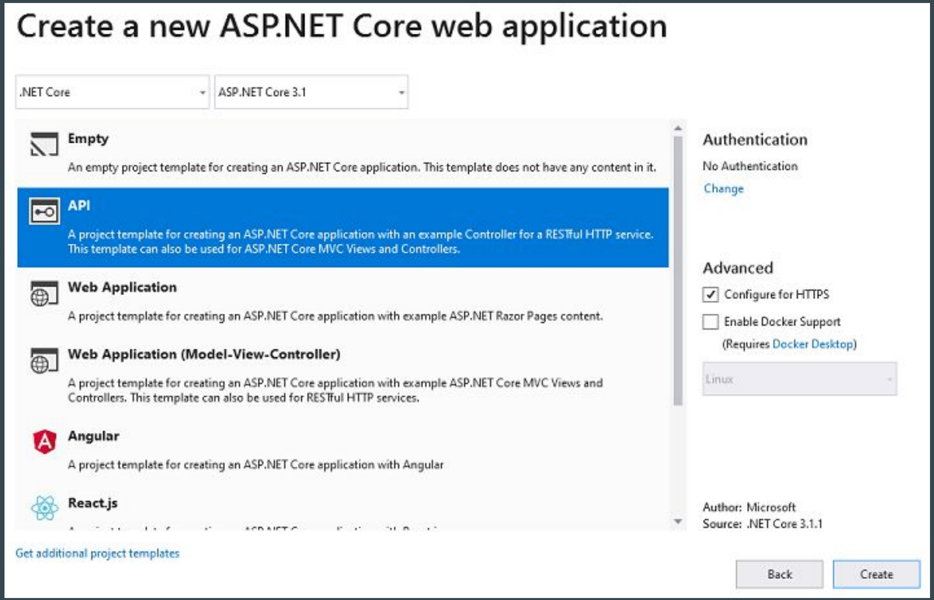
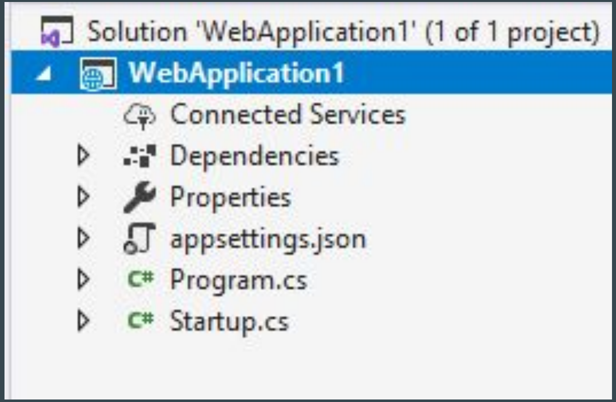
Getting Started With ASP.Net Core Web APIs in .Net 5

- Creating Simple ASP.Net Core Web API





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3 Steps : Creating A Simple ASP.Net Core Web API

1. Startup Class
   - Add services.AddControllers(); in ConfigureServices method.
   - Add app.UseRouting(); app.UseEndpoints(endpoints => { endpoints.MapControllers();}); in Configure method.
2. Add FirstController.cs in Controllers folder & Add these two attributes on the controller
   - [Route("api/[controller]")]
   - [ApiController]
3. Add Get() method in the controller class

```
public class Startup
{
    // This method gets called by the runtime. Use this method to add service
    // For more information on how to configure your application, visit https
    0 references
    public void ConfigureServices(IServiceCollection services)
    {
        services.AddControllers();
    }
    // This method gets called by the runtime. Use this method to configure t
    0 references
    public void Configure(IApplicationBuilder app, IWebHostEnvironment env)
    {
        app.UseRouting();
        app.UseEndpoints(endpoints =>
        {
            endpoints.MapControllers();
        });
    }
}
```

```
[Route("api/[controller]")]
[ApiController]
public class FirstController:ControllerBase
{
    public string Get()
    {
        return "Welcome To ManzoorTheTrainer!";
    }
}
```

Parameters Mechanism And Get Action OverLoading

```
[HttpGet("{id}")]
public IActionResult Get(int id)...
```

localhost:55629/api/department/12

{"did":12,"name":"QA","description":"Quality Assurance"}

```
[Route("[action]/{Name}")]
[HttpGet]
public IActionResult GetByName(string Name)...
```

localhost:55629/api/department/GetByName/QA

{"did":12,"name":"QA","description":"Quality Assurance"}

```
[HttpGet("getByIdAndName/{id}/{dName}")]
0 references | 0 requests | 0 exceptions
public IActionResult GetByIdAndName(int id, string dName)...
```

localhost:51294/api/Departments/getByIdAndName/1001/HR

{"did":1001,"dName":"HR","description":"Our human resurece management group"}

```
[HttpGet("getByIdAndName")]
0 references | 0 requests | 0 exceptions
public IActionResult GetByIdAndName(int id, string dName)...
```

localhost:51294/api/Departments/getByIdAndName?id=1001&dName=HR

{"did":1001,"dName":"HR","description":"Our human resurece management group"}

Controller & Action

- Controller is a class which is inherited from ControllerBase class.
- User request is handled by a controller's method called as Action
- Model is a class that describes data structures.

```
[Route("api/[controller]")]
[ApiController]
public class FirstController:ControllerBase
{
    public string Get()
    {
        return "Welcome To ManzoorTheTrainer!";
    }
}
```

```
0 references
public class Department
{
    [Key]
    0 references
    public int Did { get; set; }
    0 references
    public string DName { get; set; }
    0 references
    public string Description { get; set; }
}
```

HTTP Methods

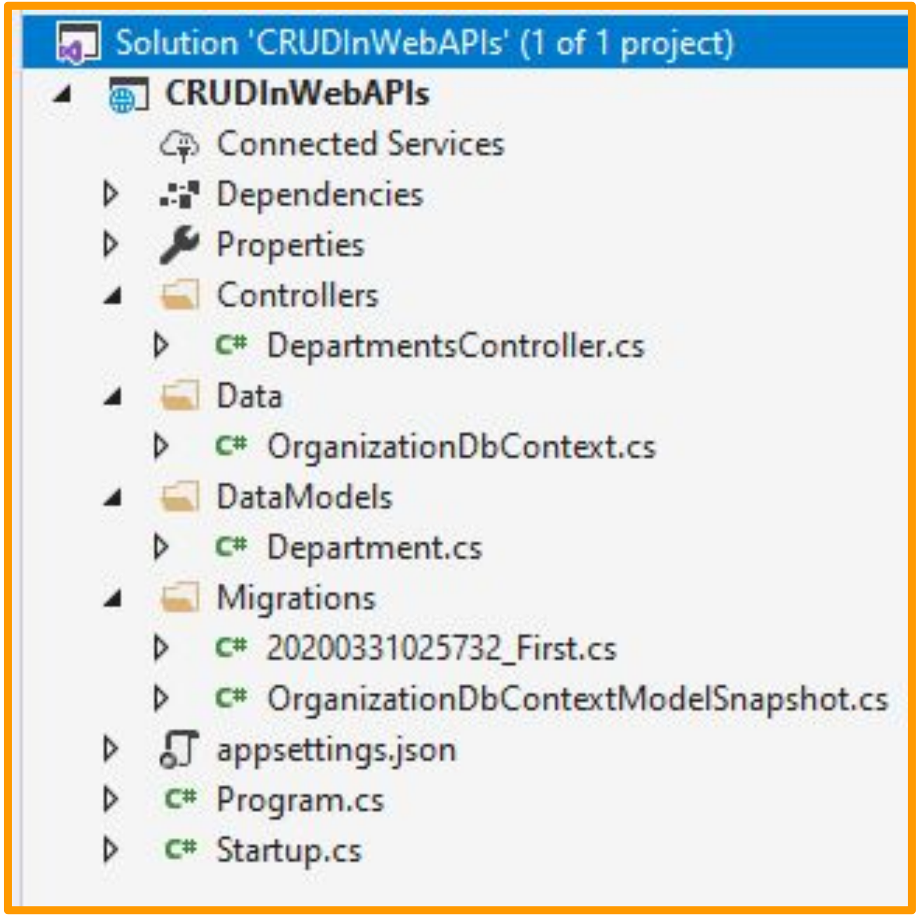
- POST → Create
- GET → Read
- PUT → Update
- DELETE → Delete
  - Eg: (GET or DELETE) URI http://manzoorthetrainer.com/courses/1

CRUD Operations Using EF Core 5.0

- Implementation of CRUD operation using EF.
    - CRUD Using EF Core in ASP.Net Core Web API
- Invoking from fiddler
    - Web Debugger Telerik (http://www.telerik.com/download/fiddler)

Create EF Core DbContext

- Start a new .Net Web Application
- Create the Directory Structure.
- Install - Nuget Packages
- Microsoft.EntityFrameworkCore.SqlServer
- Create an Entity Department
- Create a OrganizationContext Class
- OrganizationContext Class Should have DbSet<Department> properties for each entity or object.



Creating A Database - 5 Steps

1. Install - Nuget Packages
   Microsoft.EntityFrameworkCore.Tools
2. Override OnConfiguring() method in OrganizationContext with connection string.
3. Open Package Manager Console and run the below commands
4. add-migration OrganizationDb
5. update-database

```
class Department
{
    [Key]
    public int Did { get; set; }
    public string Name { get; set; }
    public string Description { get; set; }
}
```

```
class OrganizationContext:DbContext
{
    protected override void OnConfiguring(DbContextOptionsBuilder optionsBuilder)
    {
        optionsBuilder.UseSqlServer(@"Server=XXXX;Database=OrganizationDB;Trusted_Connection=True;");
    }

    public DbSet<Department> Departments { get; set; }
}
```

Package Manager Console

Package source: All   Default project: EFLatest

```
PM> add-migration OrganizationDb
To undo this action, use Remove-Migration.
PM> update-database
Applying migration '20180425065824_OrganizationDb'.
Done.
PM>
```

Dependency Injection

- .NET garbage collector performs memory cleanup for C# objects.
- But, it does not release memory for unmanaged resources.
- Unmanaged resources like Files, DbConnections, DataReaders, Printers, etc.,
- We need to implement or call a Dispose method to release unmanaged resources.
- Solution to this overhead creating object in every controller is DI.
- The EF with DI itself takes care of ensuring that DbContext objects are disposed properly so there is no need to call Dispose.

```
public class Startup
{
    // This method gets called by the runtime. Use this method t
    // For more information on how to configure your application
    0 references
    public void ConfigureServices(IServiceCollection services)
    {
        services.AddControllers();
        services.AddDbContext<OrganizationDbContext>();
    }
```

```
[Route("api/[controller]")]
[ApiController]
1 reference
public class DepartmentsController : ControllerBase
{
    OrganizationDbContext myDbContext;

    0 references
    public DepartmentsController(OrganizationDbContext myDbContext)
    {
        this.myDbContext = myDbContext;
    }
```

Creating An API - Read

1. In Startup Class
   a. Add services.AddMvc(); services.AddDbContext<OrganizationContext>(); in ConfigureServices method.
2. Add DepartmentController.cs in Controllers folder
3. Add Get() method in the controller class
4. Add [Route("api/[controller]")]
5. Add [ApiController] on Controller

```
[Route("api/[controller]")]
[ApiController]
public class DepartmentController : Controller
{
    OrganizationContext organizationContext;

    public DepartmentController(OrganizationContext _organizationContext)
    {
        organizationContext = _organizationContext;
    }
    // GET: api/<controller>
    [HttpGet]
    public IEnumerable<Department> Get()
    {
        var depts = organizationContext.Departments.ToList();
        return depts;
    }
    // GET api/<controller>/5
    [HttpGet("{id}")]
    public Department Get(int id)
    {
        Department D= organizationContext.Departments
                    .Where(x => x.Did == id).FirstOrDefault();
        return D;
    }
```

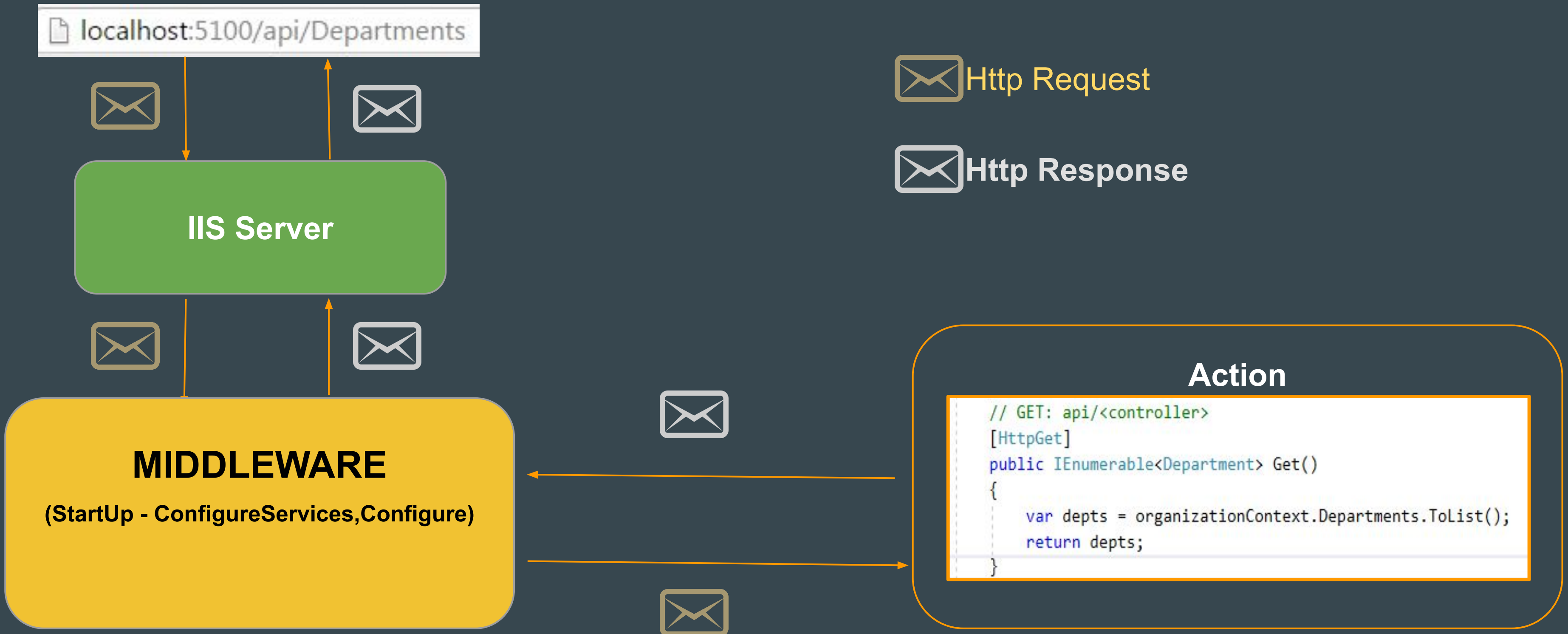
Insert, Update & Delete

1. Add Post(), Put() & Delete method in the controller class.
2. Let’s Use Fiddler to test our Web API

Note: While Performing Insert and Update, don't forget to set content-type in request header.
Content-Type: application/json

```
[HttpPost]
public Department Post(Department D)
{
        organizationContext.Add<Department>(D);
        organizationContext.SaveChanges();
        return D;
}
//PUT api/<controller>/5
[HttpPut]
public string Put( Department D)
{
        organizationContext.Update<Department>(D);
        organizationContext.SaveChanges();
        return "Record Updated Successfully";
}
// DELETE api/<controller>/5
[HttpDelete("{id}")]
public string Delete(int id)
{
    Department D = organizationContext.Departments
        .Where(x => x.Did == id).FirstOrDefault();
        organizationContext.Remove<Department>(D);
        organizationContext.SaveChanges();
        return "Record Deleted Successfully";
}
```

ASP.Net Core Web API Life Cycle



Http Request

- Content(body)
- Headers
- Method (Get,Post,Put or Delete)
- RequestUri

HTTP Response (Data + Response Code)

- Response Code
  - 200 - HttpStatusCode.OK
  - 201 - HttpStatusCode.Created
  - 204 - HttpStatusCode.NoContent
  - 400 - HttpStatusCode.BadRequest
  - 401 - HttpStatusCode.Unauthorized
  - 403 - HttpStatusCode.Forbidden
  - 404 - HttpStatusCode.NotFound
  - 500 - HttpStatusCode.InternalServerError
  - For more http://www.w3schools.com/tags/ref_httpmessages.asp

Creating Response - Get

- Say I am looking for a department as D.
    - OK();
    - NotFound();
- Return type can be
    - ActionResult<Department>
    - IActionResult

```
[HttpGet()]
public ActionResult<Department> Get(int id)
{
    Department D= organizationContext.Departments
                    .Where(x => x.Did == id).FirstOrDefault();
    if (D != null)
    {
        return Ok(D);// Data + 200 (Response Code)
    }
    else
    {
        return NotFound(); //404 (Response Code)
    }
}
```

```
[HttpGet()]
public IActionResult Get(int id)
{
    Department D= organizationContext.Departments
                    .Where(x => x.Did == id).FirstOrDefault();
    if (D != null)
    {
        return Ok(D);// Data + 200 (Response Code)
    }
    else
    {
        return NotFound(); //404 (Response Code)
    }
}
```

Creating Response - Post

- Say I am inserting a Department
  - CreatedAtAction()
  - BadRequest()

Note: CreatedAtAction will return object created along with the uri to access that object. (Location)

```csharp
[HttpPost]
public IActionResult Post(Department D)
{
    if (ModelState.IsValid)
    {
        organizationContext.Add<Department>(D);
        organizationContext.SaveChanges();
        return CreatedAtAction("Get",new { id = D.Did }, D);
    }
    else
    {
        return BadRequest(ModelState);
    }
}
```

Transport

Location: http://localhost:55629/api/Department/18

Transfer-Encoding: chunked

Creating Response - Put

- Say I am updating a department
  - BadRequest()
  - NotFound()
  - NoContent()
  - OK()

```
[HttpPut]
0 references | 0 requests | 0 exceptions
public IActionResult Put(Department D)
{
    if (ModelState.IsValid)
    {
        var Dept = dbContext.Departments
                        .Where(x => x.Did == D.Did)
                        .AsNoTracking().FirstOrDefault();
        if (Dept != null)
        {
            dbContext.Update(D);
            dbContext.SaveChanges();
            return NoContent();//or Ok(D);
        }
        else
        {
            return NotFound();
        }
    }
    else
    {
        return BadRequest(ModelState);
    }
}
```

Creating Response - Delete

- Say I am deleting a Department
  - NotFound()
  - NoContent()
  - OK()

```
[HttpDelete("{id}")]
public IActionResult Delete(int id)
{
    Department D = organizationContext.Departments
                    .Where(x => x.Did == id).FirstOrDefault();
    if (D != null)
    {
        organizationContext.Remove<Department>(D);
        organizationContext.SaveChanges();
        return NoContent();//  or return Ok(D);
    }
    else
    {
        return NotFound();
    }
}
```

Asynchronous Actions & Scaffolding

- async,Task and await
- Adding Employee entity and Scaffolding EmployeesController

```
[Route("api/[controller]")]
[ApiController]
1 reference
public class EmployeesController : ControllerBase
{
    private readonly OrganizationDbContext _context;
    0 references | 0 exceptions
    public EmployeesController(OrganizationDbContext context)
    {
        _context = context;
    }

    [HttpGet]
    0 references | 0 requests | 0 exceptions
    public async Task<ActionResult<IEnumerable<Employee>>> GetEmployees()
    {
        var Emps = await _context.Employees.ToListAsync();
        return Ok(Emps);
    }

    [HttpGet("{id}")]
    0 references | 0 requests | 0 exceptions
    public async Task<ActionResult<Employee>> GetEmployee(int id)
    {
        var employee = await _context.Employees.FindAsync(id);
```

Server Side Validations

```
public class Employee
{
    [Key]
    3 references
    public int Eid { get; set; }
    [Required]
    0 references
    public string Gender { get; set; }
    [Required]
    1 reference
    public string FirstName { get; set; }
    1 reference
    public string LastName { get; set; }
    [NotMapped]
    0 references
    public string FullName { get { return FirstName + " " + LastName; } }
    [UniqueEmail]
    [EmailAddress]
    1 reference
    public string Email { get; set; }
    [NotMapped]
    [Required,EmailAddress,Compare("Email")]
    0 references
    public string ConfirmEmail { get; set; }
    [Range(10000,100000)]
    0 references
    public double Salary { get; set; }
    [Url]
    0 references
    public string LinkedInProfile { get; set; }
    [ForeignKey("Department")]
    1 reference
    public int Did { get; set; }
    0 references
    public Department Department { get; set; }
}
```

Custom Validators - 3 Steps

1. Create a class for custom validator say **UniqueEmailAttribute** which inherits **ValidationAttribute**.
2. Override the method **IsValid()** with validation logic which takes input as **value** and returns **ValidationResult**
3. Then apply [**UniqueEmail**] attribute on email field where ever is required.

```
public class UniqueEmailAttribute:ValidationAttribute
{
    protected override ValidationResult IsValid(object value,
                                   ValidationContext validationContext)
    {
        SchoolDbContext schoolDbContext = new SchoolDbContext();
        if (value != null)
        {
            int count = schoolDbContext
                                .Students
                                .Where(x => x.Email == value.ToString())
                                .Count();
            if (count == 0)
                return ValidationResult.Success;
            else
                return new ValidationResult("This Email already exist!");
        }
        else
        {
            return new ValidationResult("Email is required!");
        }
    }
}
```

```
[Table("Student")]
public class Student
{
    [Key]
    public int StudentId { get; set; }

    [UniqueEmail] //Custom Validator
    public string Email { get; set; }
}
```

Solution To Circular Reference Problem

Approaches - 1:

Write specific Select Lambda expression

```
[HttpGet]
0 references | 0 requests | 0 exceptions
public async Task<ActionResult<IEnumerable<Employee>>> GetEmployees()
{
    var Emps = await _context.Employees.Include(x => x.Department)
                .Select(x => new Employee
                {
                    Eid = x.Eid,
                    Name = x.Name,
                    Gender = x.Gender,
                    Did = x.Did,
                    Department = x.Department
                })
                .ToListAsync();
    return Ok(Emps);
}
```

Use Lamda Expression

```
[HttpGet]
0 references | 0 requests | 0 exceptions
public async Task<IActionResult> Get()
{
    var Depts = await dbContext.Departments.Include(x => x.Employees)
        .Select(x => new Department
        {
            Did = x.Did,
            DName = x.DName,
            Description = x.Description,
            Employees = x.Employees.Select(y => new Employee
            {
                Eid = y.Eid,
                Name = y.Name,
                Gender = y.Gender
            })
        })
        .ToListAsync();
    return Ok(Depts);
}
```

Lamda Expression

Solution To Circular Reference Problem

Approaches - 2:

1. Add Newtonsoft.Json from Manage nuget packages.
2. Serialize the Depts object to JSON string using JsonConvert by ignoring reference looping in serialization setting

```
[HttpGet]
0 references | 0 requests | 0 exceptions
public async Task<IActionResult> Get()
{
    var Depts = await dbContext.Departments
                        .Include(x => x.Employees).ToListAsync();

    if (Depts.Count != 0)
    {
        var jsonResult = JsonConvert.SerializeObject(Depts,
            Formatting.None,
            new JsonSerializerSettings()
            {
                ReferenceLoopHandling = ReferenceLoopHandling.Ignore
            });
        return Ok(jsonResult);
    }
    else
        return NotFound();
}
```

Response - Get As XML

- Default json: String, Specific, ActionResult<Type>, IActionResult
- For XML
  - XMLInstall the package Microsoft.AspNetCore.Mvc.Formatters.Xml
  - Add xml format in service services.AddControllers() .AddXmlSerializerFormatters() .AddXmlDataContractSerializerFormatters();
  - Add Filter on the controller [Produces("application/xml")]

```
services.AddMvc()
        .AddXmlSerializerFormatters()
        .AddXmlDataContractSerializerFormatters();
```

```
[Produces("application/xml")]
[Route("api/[controller]")]
[ApiController]
public class DepartmentController : Controller
{
    OrganizationContext organizationContext;
```

Swagger Or OpenAPI - 3 Steps

Swagger generates documentation for Web API so that developer can understand & consume Web API.

1. Add Package NSwag.AspNetCore
2. Add services.AddSwaggerDocument(); in as service in startup.cs
3. Use app.UseOpenApi(); app.UseSwaggerUi3(); In configure of startup.cs
4. Access Doc from http://localhost:61964/swagger/index.html

```
public void ConfigureServices(IServiceCollection services)
{
    services.AddMvc();
    services.AddDbContext<OrganizationDbContext>();
    services.AddSwaggerDocument();
}

// This method gets called by the runtime. Use this method
0 references | 0 exceptions
public void Configure(IApplicationBuilder app, IHostingEnv
{
    app.UseDeveloperExceptionPage();
    app.UseOpenApi();
    app.UseSwaggerUi3();
    app.UseMvc();
}
```

Exception Handling

Action Level

```
[HttpGet("{id}")]
public IActionResult Get(int id)
{
    try
    {
        Department D = organizationContext
                        .Departments
                        .Where(x => x.Did == id)
                        .FirstOrDefault();
        if (D != null)...
        else...
    }
    catch (Exception E)
    {
        //Write to exception Logger
        return StatusCode(500, E.Message);
    }
}
```

Exception Handling Globally - Custom Filter

```
public class MyExceptionFilter : IExceptionFilter
{
    public void OnException(ExceptionContext context)
    {
        context.ExceptionHandled = true;
        HttpResponse response = context.HttpContext.Response;
        response.StatusCode = 500;
        response.ContentType = "application/json";
        context.Result = new ObjectResult(context.Exception.Message);
    }
}
```

```
public void ConfigureServices(IServiceCollection services)
{
    services.AddMvc(config =>
    {
        config.Filters.Add(typeof(MyExceptionFilter));
    });
}
```

```
[HttpGet("{id}")]
public IActionResult Get(int id)
{
        Department D = organizationContext
                        .Departments
                        .Where(x => x.Did == id)
                        .FirstOrDefault();
        if (D != null)...
        else...
}
```

Here We Need Not To Write Try..Catch

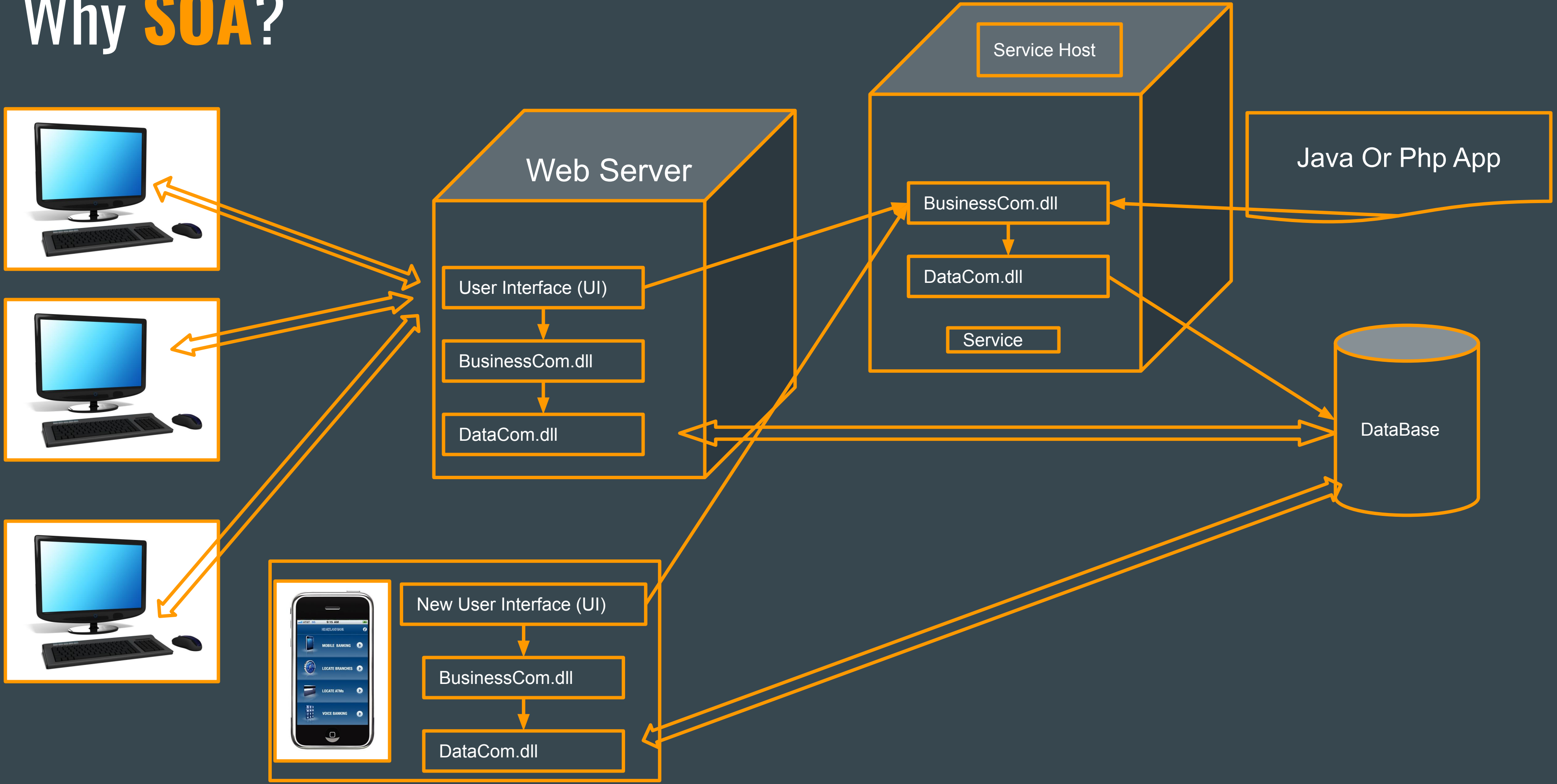
Exception Handling Globally - In Middleware


```
public void Configure(IApplicationBuilder app, IHostingEnvironment env)
{
    app.UseExceptionHandler(
        options =>
        {
            options.Run(async context =>
            {
                context.Response.StatusCode = 500;//Internal Server Error
                context.Response.ContentType = "application/json";
                var ex = context.Features.Get<IExceptionHandlerFeature>();
                if (ex != null)
                {
                    await context.Response.WriteAsync(ex.Error.Message);
                }
            });
        }
        );
    app.UseMvc();
}
```

```
[HttpGet("{id}")]
public IActionResult Get(int id)
{
        Department D = organizationContext
                            .Departments
                            .Where(x => x.Did == id)
                            .FirstOrDefault();
        if (D != null)...
        else...
}
```

Here We Need Not To Write Try..Catch

Why SOA?



Service Host
Web Server
Java Or Php App
BusinessCom.dll
DataCom.dll
Service
User Interface (UI)
BusinessCom.dll
DataCom.dll
DataBase
New User Interface (UI)
BusinessCom.dll
DataCom.dll

What is **SOA**?

- It is a reusable component on the network.
- It is the collection of services on a network that communicate with one another.
- For example, verifying a credit card transaction or processing a purchase order.
- Loosely coupled (meaning that an application doesn't have to know the technical details of another application in order to talk to it).
- Services are well-defined, platform-independent interfaces, and are reusable.

How To Achieve SOA?

Few famous ways of achieving SOA

- .NET Remoting
- Web services
- WCF
- **RESTful Services Web APIs**

OLD

NEXT

That's why REST?

- Service for any device with front-end
  - Easy
  - Simple
  - Light weight
  - All features of HTTP
  - ReST-ful (Representational State Transfer) Services fulfill all the above needs.

What is REST?

"Architectural Styles and the Design of Network-based Software Architectures" was initially proposed by Roy Thomas Fielding in his 2000 Ph.D. dissertation.

What Is RESTful Web APIs?

- Web service APIs that adhere to the REST architectural constraints.
- HTTP-based RESTful APIs Clients should know:
  - Base URI, such as http://manzoorthetrainer.com/courses/
  - An Internet media type for the data. This is often JSON but can be any other valid Internet media type (e.g., XML, images, etc.)
  - Standard HTTP methods (e.g., GET, PUT, POST, or DELETE)
- Few famous Web APIs are Google Maps, Twitter, YouTube, Flickr, Facebook, Amazon Product, Advertising, etc.,

Security In Web API

- Authentication & Authorization with ASP.Net Identity as user management
  - Cookie Based (Default)
  - Token Based Authentication (JWT)

ASP.Net Identity - Authentication

- Checking the genuinity of a user or client is called as authentication.
- It is normally achieved with the help of username and password called as forms authentication for web apps.
- In our early days .Net Framework we used Membership classes or providers to achieve authentication.
- The ASP.NET Identity system is a replacement of Membership systems which is specially designed to support social logins for building modern applications for the web, phone, or tablet.
- In Microsoft.AspNetCore.Identity we have mainly three classes using which we implement authentication and authorization i.e., RoleManager, UserManager and SignInManager

Securing Web API

1. Secure Access [Authorize]
   a. Action Level
   b. Controller Level
   c. Application Level
2. Anonymous Access [AllowAnonymous]
3. Send Response Code as 403 on OnRedirectToLogin event

```
[Authorize]
[HttpGet]
0 references | 0 requests | 0 exceptions
public async Task<IActionResult> Get()...
```

```
[Authorize]
[Route("api/[controller]")]
[ApiController]
1 reference
public class DepartmentsController : Controller
{
```

```
0 references | 0 exceptions
public void ConfigureServices(IServiceCollection services)
{
    services.AddMvc(x => x.Filters.Add(new AuthorizeFilter()));
```

```
services.ConfigureApplicationCookie(opt =>
            {
                opt.Events = new CookieAuthenticationEvents()
                {
                    //Authentication
                    OnRedirectToLogin = redirectContext =>
                    {
                        redirectContext.HttpContext.Response.StatusCode = 403;
                        return Task.CompletedTask;
                    }
                };
            });
```

Initial Setup - 5 Steps

1. Add the nuget package Microsoft.AspNetCore.Identity.EntityFrameworkCore
2. Inherit application's DbContext say OrganizationDbContext from IdentityDbContext.
3. Add Migration, Update-Database
4. Inject application's DbContext object and Identity object using service collection object in ConfigureServices Method of StartUp Class.
5. Add authentication configuration in HTTP pipeline using app object in Configure method of StartUp classes

```
10 references
public class OrganizationDbContext:IdentityDbContext
{
    0 references | 0 exceptions
    protected override void OnConfiguring(DbContextOptionsBui
    {
        base.OnConfiguring(optionsBuilder);
        optionsBuilder.UseSqlServer(@"Server=DESKTOP-084LAFN;
    }
```

```
0 references | 0 exceptions
public void ConfigureServices(IServiceCollection services)
{
    services.AddMvc(x => x.Filters.Add(new AuthorizeFilter()));
    services.AddDbContext<OrganizationDbContext>();
    services.AddIdentity<IdentityUser, IdentityRole>()
        .AddEntityFrameworkStores<OrganizationDbContext>()
        .AddDefaultTokenProviders();
```

```
public void Configure(IApplicationBuilder app, IWebHostEnvironment env)
{
    InitialSettings
    app.UseRouting();
    app.UseAuthentication();
    app.UseAuthorization();
    app.UseEndpoints(endpoints =>
    {
        endpoints.MapControllers();
    });
}
```

User Registration - 4 Steps

1. Create RegisterViewModel
2. Create UserManager And SignInManager objects.
3. Create a Post action to handle registration request.

```
[AllowAnonymous]
[Route("api/[controller]")]
[ApiController]
1 reference
public class AccountController : ControllerBase
{
    UserManager<IdentityUser> userManager;
    SignInManager<IdentityUser> signInManager;

    0 references | 0 exceptions
    public AccountController(SignInManager<IdentityUser> _signInManager,
        UserManager<IdentityUser> _userManager)
    {
        signInManager = _signInManager;
        userManager = _userManager;
    }
```

```
[HttpPost("register")]
0 references | 0 requests | 0 exceptions
public async Task<IActionResult> Register(RegisterViewModel model)
{
    if (ModelState.IsValid)
    {
        var user = new IdentityUser()
        {
            UserName = model.UserName,
            Email = model.Email
        };
        var userResult = await userManager.CreateAsync(user, model.Password);
        if (userResult.Succeeded)
        {
            return Ok(user);
        }
    }
    return BadRequest(ModelState.Values);
}
```

SignIn And SignOut - 7 Steps

1. Create SignInViewModel
2. Create UserManager And SignInManager objects.
3. Create an action to handle SignIn and SignOut request.

Note: After SignIn we get a cookie & we need to pass that cookie in the header for secure access of any actions.

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```
[AllowAnonymous]
[Route("api/[controller]")]
[ApiController]
1 reference
public class AccountController : ControllerBase
{
    UserManager<IdentityUser> userManager;
    SignInManager<IdentityUser> signInManager;

    0 references | 0 exceptions
    public AccountController(SignInManager<IdentityUser> _signInManager,
        UserManager<IdentityUser> _userManager)
    {
        signInManager = _signInManager;
        userManager = _userManager;
    }
```

```
[HttpPost("signIn")]
0 references | 0 requests | 0 exceptions
public async Task<IActionResult> SignIn(SignInViewModel model)
{
    if (ModelState.IsValid)
    {
        var signInResult = await signInManager
            .PasswordSignInAsync(model.UserName, model.Password, false, false);
        if (signInResult.Succeeded)
        {
            return Ok();
        }
    }
    return BadRequest(ModelState);
}
[HttpPost("signOut")]
0 references | 0 requests | 0 exceptions
public async Task<IActionResult> SignOut()
{
    await signInManager.SignOutAsync();
    return NoContent();
}
```

Authorization - 4 Steps

1. Create Roles in Db Say 'Admin' and 'User'.
2. Assign Role to a user from Db.
3. Add role to authorize attribute [Authorize(Roles = "Admin")]
4. At last we will see how to assign default role as 'User' at the time of registration.

Note : Send Response Code as 401 on OnRedirectToAccessDenied event & set cookie expiration time.

```
[HttpPost("register")]
0 references | 0 requests | 0 exceptions
public async Task<IActionResult> Register(RegisterViewModel model)
{
    if (ModelState.IsValid)
    {
        var user = new IdentityUser()...;
        var userResult = await userManager.CreateAsync(user, model.Password);
        if (userResult.Succeeded)
        {
            var roleResult = await userManager.AddToRoleAsync(user, "User");
            if (roleResult.Succeeded)
            {
                return Ok(user);
            }
        }
    }
    return BadRequest(ModelState.Values);
}
```

```
services.ConfigureApplicationCookie(opt =>
{
    opt.ExpireTimeSpan = new TimeSpan(0, 0, 30);
    opt.Events = new CookieAuthenticationEvents
    {
        OnRedirectToLogin = redirectContext =>
        {
            redirectContext.HttpContext.Response.StatusCode = 401;
            return Task.CompletedTask;
        },
        OnRedirectToAccessDenied = redirectContext =>
        {
            redirectContext.HttpContext.Response.StatusCode = 401;
            return Task.CompletedTask;
        }
    };
});
```

Few Security Concepts

- Principal
  - Identity
  - Role
- Claims
- https://github.com/microsoft/referencesource/tree/master/mscorlib/system/security
- We need
  - UserId
  - UserName
  - Role

Why JWT & Why not Cookies for Authentication?

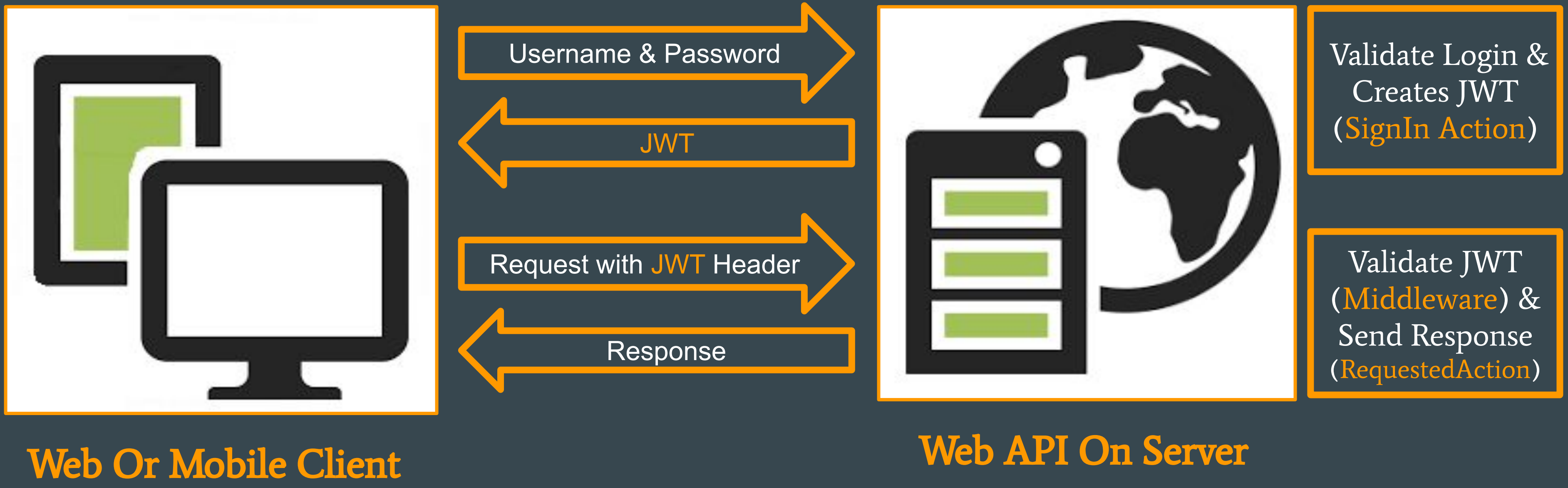
1. It is stateless.
2. JSON parsers are common in most programming languages.
3. Client-side support on multiple platforms, especially mobile.

Note: We need to add these packages Microsoft.IdentityModel.Tokens.Jwt & Microsoft.AspNetCore.Authentication.JwtBearer

What is JWT (JSON Web Tokens)? (https://jwt.io)

- JSON Web Token (JWT) is a way of securely transmitting information between client and server as a JSON object.
- This information is digitally signed using a secret key (HMAC algorithm).
- Basically it is used for Authorization and Information Exchange.
- Structure Of JWT
  a. Header (ALGORITHM & TOKEN TYPE)
  b. PayLoad(Data)
  c. Sign (Header+Payload+Key)
- Eg: xxxxxxxxxxx.yyyyyyyyyyyyyyy.zzzzzzzzzzzzzzzz

How JWT Works? (Authentication & Authorization)



Create JWT in SignIn Action (Steps - 5)

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```
[HttpPost("signIn")]
0 references | 0 requests | 0 exceptions
public async Task<IActionResult> SignIn(SignInViewModel model)
{
    Find user and it's roles after successful login
    //Step - 1: Create IdentityClaims
    IdentityOptions identityOptions = new IdentityOptions();
    var claims = new Claim[]
    {
            new Claim("userId",user.Id),
            new Claim(identityOptions.ClaimsIdentity.UserNameClaimType,user.UserName),
            new Claim(identityOptions.ClaimsIdentity.RoleClaimType, roles[0])
    };

    //Step - 2: Create signingKey from Secretkey
    var signingKey = new SymmetricSecurityKey(Encoding.UTF8.GetBytes("this-is-my-secret-key"));

    //Step -3: Create signingCredentials from signingKey with HMAC algorithm
    var signingCredentials = new SigningCredentials(signingKey, SecurityAlgorithms.HmacSha256);

    //Step - 4: Create JWT with signingCredentials, IdentityClaims & expire duration.
    var jwt = new JwtSecurityToken(signingCredentials: signingCredentials,
                                    expires: DateTime.Now.AddMinutes(30), claims: claims);

    //Step - 5: Finally write the token as response with OK().
    return Ok(new JwtSecurityTokenHandler().WriteToken(jwt));
}
```

Validate JWT in Middleware (Steps - 5)


```
//Step-1: Create signingKey from Secretkey
var signingKey = new SymmetricSecurityKey(Encoding.UTF8.GetBytes("this-is-my-secret-key"));

//Step-2:Create Validation Parameters using signingKey
var tokenValidationParameters = new TokenValidationParameters()
{
    IssuerSigningKey = signingKey,
    ValidateIssuer = false,
    ValidateAudience = false
};

        //Step-3: Set Authentication Type as JwtBearer
services.AddAuthentication(auth =>
        {
            auth.DefaultAuthenticateScheme = JwtBearerDefaults.AuthenticationScheme;
        })
        //Step-4: Set Validation Parameter created above
        .AddJwtBearer(jwt =>
        {
            jwt.TokenValidationParameters = tokenValidationParameters;
        });
```

Note:

```
[Authorize(AuthenticationSchemes =JwtBearerDefaults.AuthenticationScheme,Roles ="Admin")]
[Route("api/[controller]")]
[ApiController]
1 reference
public class DepartmentsController : Controller
{
```

Thanks